RACHEL VAN DRUNEN

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EDUCATION

MD Anderson Cancer Center UT Health Houston Graduate School of Biomedical Sciences
Ph.D., Neuroscience **August 2024**

Trinity University
Bachelor of Science (B.Sc.), Neuroscience **May 2018**

RESEARCH

Department of Neuroscience, Brown University **July 2024 – Present**
Postdoctoral Researcher, Advisor: Dr. Alexander Fleischmann
Molecular and functional circuitry underlying the sensory processing of the mouse olfactory cortex.

Department of Biomolecular Sciences, The Weizmann Institute of Science **January 2024 – June 2024**
Fulbright Fellow, Advisor: Dr. Gad Asher
Analyzed single cell RNA sequencing data from the lung tissue of a mouse with a liver specific BMAL1 and Hif1a double knockout.

Metabolic Diseases and Disorders, MD Anderson UTHealth Science Center in Houston **May 2019 – December 2023**
Ph.D. Candidate, Advisor: Dr. Kristin Eckel-Mahan
Dissertation Title: The Rhythmic Role of BMAL1 in the Paraventricular Nucleus of the Hypothalamus for Diurnal Energy Balance.

Department of Engineering, Trinity University **May 2017 – May 2018**
Undergraduate research student, Advisor: Dr. Dany J. Munoz-Pinto
Fabricated and tested mIPNs for their ability to recapitulate the *in-vivo* extracellular environment of the mouse brain.

Department of Epidemiology, MD Anderson Cancer Center **Summer 2016**
Summer research internship, Advisor: Dr. Michelle Hildebrandt
Analyzed RNA sequencing data from doxorubicin (DOX) treated cardiomyocytes to identify gene expression changes in the cardiac contraction pathway.

Regenerative Medicine Research Lab, Texas Heart Institute **Summers 2014 & 2015**
Summer research student, Advisor: Dr. Doris Taylor
Worked with a small team to develop a non-thrombogenic device using reendothelized porcine bladder with the long-term goal of mitigating the adverse side effects of the Fontan procedure.

PUBLICATIONS

1. Van Drunen, R., Fekry, B., Haichao, W., Dai, Y., Bravo Santos, R., Shivshankar, S., Zhao, Z., Jiaqian, W., Qingchun, T. and Eckel-Mahan, K. (2024) **Cell type-specific regulation of the circadian clock by BMAL1 in the paraventricular nucleus: implications for regulation of energy balance.** *Cell Reports*, vol 43(7).
2. Dandavate, V., N. Bolshette, Van Drunen, R., G. Manella, H. Bueno-Levy, M. Zerbib, I. Kawano, M. Golik, Y. Adamovich, and G. Asher. (2024). **Hepatic BMAL1 and HIF1alpha regulate a time-dependent hypoxic response and prevent hepatopulmonary-like syndrome.** *Cell Metabolism*, **36**(9): p. 2038-2053.e5.

3. Gao, Z., R.B. Santos, J. Rupert, Van Drunen, R., Y. Yu, K. Eckel-Mahan, and M.G. Kolonin. (2024). **Endothelial-specific telomerase inactivation causes telomere-independent cell senescence and organ dysfunction characteristic of aging**. Aging Cell, 23(6): p. e14138.
4. Sopariwala, D.H., Rios, A.S., Pei, G., Roy, A., Tomaz da Silva, M., Thi Thu Nguyen, H., Saley, A., Van Drunen, R., Kralli, A., Mahan, K. and Zhao, Z. (2023). **Innately expressed estrogen-related receptors in the skeletal muscle are indispensable for exercise fitness**. *The FASEB Journal*, *37*(2), p.e22727.
5. Fekry, B., Ribas-Latre, A., Van Drunen, R., Bravo Santos, R., Shivshankar, S., Dai, Y., Zhao, Z., Yoo, S.H., Chen, Z., Sun, K., Sladek, F.M. and Eckel-Mahan, K. (2022). **Hepatic circadian and differentiation factors control liver susceptibility for fatty liver disease and tumorigenesis**. *The FASEB Journal*, 36(9), p.e22482.
6. Van Drunen, R. and Eckel-Mahan, K. (2021) **Circadian Rhythms of the Hypothalamus: from Function to Physiology.** *Clocks & Sleep* ;3(1):189–226.
7. Kim, E. R., Xu, Y., Cassidy, R. M., Lu, Y., Yang, Y., Tian, J., Li, D. P., Van Drunen, R., Ribas-Latre, A., Cai, Z. L., Xue, M., Arenkiel, B. R., Eckel-Mahan, K., Xu, Y., & Tong, Q. (2020). **Paraventricular hypothalamus mediates diurnal rhythm of metabolism**. *Nature communications*, *11*(1), 3794.
8. Jimenez-Vergara A.C., Van Drunen R., Cagle T., Munoz-Pinto D.J. (2020) **Modeling the Effects of Hyaluronic Acid Degradation on the Regulation of Human Astrocyte Phenotype Using Multicomponent Interpenetrating Polymer Networks (mIPNs).** *Scientific Reports,* 10:20734.
9. Van Drunen R., Jimenez-Vergara A.C., Tsai E.H., Tchen R., Cagle T., Agee A.B., Roberts J., Steele J.M., Munoz-Pinto D.J. (2019) **Collagen based multicomponent interpenetrating networks as promising scaffolds for 3D culture of human neural stem cells, human astrocytes, and human microglia**. *ACS Applied Bio Materials*, 2 (3), 975-980.

AWARDS AND SCHOLARSHIPS

2023 **President's Research Scholarship,** MD Anderson UTHealth Houston GSBS
2023 **Fulbright US Student Program Open Study/Research Award,** Fulbright US Student Program
2022 **First Place UTHealth Neuroscience Program Retreat Data Poster Competition,** MD Anderson UTHealth Houston GSBS
2022 **Merit Award SRBR Annual Conference**, Society for Research in Biological Rhythms
2022 **Neuroscience Program Travel Award**, MD Anderson UTHealth Houston GSBS
2021 **Dee S. and Patricia Osborne Endowed Scholarship in the Neurosciences**, Neuroscience Research Symposium
2021 **GSBS Travel Award**, MD Anderson UTHealth Houston GSBS
2021 **John J Kopchick Research Award**, MD Anderson UTHealth Graduate School of Biomedical Sciences
2021 **F31 Ruth L. Kirschstein Predoctoral Individual National Research Service Award**, NIDDK National Institute of Health
2021 **Spring Neuroscience Student Workshop Presentation Award**, MD Anderson UTHealth Houston GSBS
2021 **Second Place, UTHealth Neuroscience Research Center Brain Awareness Week Video Contest**, Neuroscience Research Center
2021 **Second Place, UTHealth Neuroscience Program Retreat Data Blitz Competition**, MD Anderson UTHealth Houston GSSB
2021 **Second Place, McGovern Medical School Research Retreat 2021 Poster Competition**, MD Anderson UTHealth Houston GSBS
2020 **Merit Award SRBR Annual Conference (virtual),** Society for Research in Biological Rhythms
2019 **Third Place, MD Anderson UTHealth GSBS 2019 Annual Report Scientific Writing Competition,** MD Anderson UTHealth Houston GSBS
2017 **Murchison Undergraduate Research Fellowship**, Trinity University
2016 **Summer Undergraduate Research Program,** MD Anderson Cancer Center
2014 **Murchison Undergraduate Scholarship,** Trinity University

LEADERSHIP AND TEACHING EXPEREINCES

February 2022	Neuroscience and Genetics & Epigenetics Art Fair Organizer
May 2021	Mic Carrier, Cold Spring Harbor Laboratory 85th Symposium for Biological Time Keeping
2020-2021	Healthcare Innovation Challenge Advisory Committee
2019-2023	Graduate student peer mentor
2019-2022	Neuroscience Student Journal Club
2019-2021	Neuroscience Program Student Council
2019-2020	Healthcare Innovation Challenge Executive Committee Member
Fall 2019	Neuroanatomy Teaching Assistant

SELECTED ORAL PRESENTATIONS

Van Drunen, R., A., Fekry, B., Bravo Santos, R., Haichao, W., Yulin, D., Jiaqian, W., Tong, Q., and Eckel-Mahan, K. **Transcriptional regulation of the PVN clock by BMAL1.** (Shot Talk Presentation) Texas Society for Circadian Biology and Medicine (TSCBM) Meeting, Houston, TX, 2022.

Van Drunen, R., A., Fekry, B., Bravo Santos, R., Haichao, W., Yulin, D., Jiaqian, W., Tong, Q., and Eckel-Mahan, K. **The Rhythmic Role of the Paraventricular Nucleus of the Hypothalamus in Diurnal Energy Balance.** (Shot Talk Presentation) Neuronal Control of Appetite Keystone Symposia, Banff Springs, CA, Canada 2022.

Van Drunen, R., A., Fekry, B., Bravo Santos, R., Haichao, W., Yulin, D., Jiaqian, W., Tong, Q., and Eckel-Mahan, K. **The Rhythmic Role of the Paraventricular Nucleus of the Hypothalamus in Diurnal Energy Balance.** (Short Talk Presentation) Society for Research in Biological Rhythms, Amelia Island, FL 2022.

Van Drunen, R., A., Fekry, B., Bravo Santos, R., Haichao, W., Yulin, D., Jiaqian, W., Tong, Q., and Eckel-Mahan, K. **The Role of BMAL1 in the Rhythmic Paraventricular Nucleus of the Hypothalamus in Diurnal Energy Balance**. (Short Talk Presentation) McGovern Medical School Research Retreat 2021.

Van Drunen, R., Tsai, E., Jimenez-Vergara, A., and Munoz-Pinto, D.: **Tuning and characterization of multi interpenetrating networks of varying chemical composition to create a 3D model for neural trauma and neurodegenerative diseases**. (Short Talk Presentation) Gulf Coast Undergraduate Research Symposium, Houston, TX, 2017.

SELECTED POSTER PRESENTATIONS

Van Drunen, R., A., Fekry, B., Bravo Santos, R., Haichao, W., Yulin, D., Jiaqian, W., Tong, Q., and Eckel-Mahan, K. **The Rhythmic Role of BMAL1 in the Paraventricular Nucleus of the Hypothalamus in Diurnal Energy Balance.** MD Anderson UTHealth GSBS Neuroscience Research Retreat, Houston, TX 2022.

Van Drunen, R., A., Fekry, B., Bravo Santos, R., Haichao, W., Yulin, D., Jiaqian, W., Tong, Q., and Eckel-Mahan, K. **The Rhythmic Role of the Paraventricular Nucleus of the Hypothalamus in Diurnal Energy Balance.** 27th Annual Neuroscience Research Poster Session, Houston, TX 2021.

Van Drunen, R., A., Fekry, B., Bravo Santos, R., Haichao, W., Yulin, D., Jiaqian, W., Tong, Q., and Eckel-Mahan, K. **The Rhythmic Role of the Paraventricular Nucleus of the Hypothalamus in Diurnal Energy Balance.** Cold Spring Harbor Laboratory 85th Symposium for Biological Time Keeping 2021.

Van Drunen, R., Ribas Latre, A., Fekry, B., Tamin Mohamed, Y.,Tong, Q., and Eckel-Mahan, K. **A novel mechanism for rhythmic regulation of *Bmal1* in the PVN.** Society of Research for Biological Rhythms 2020.

Van Drunen, R., Tsai, E., Jimenez-Vergara, A., Munoz-Pinto, D.: **Tuning and characterization of multi interpenetrating networks of varying chemical composition to create a 3D model for neural trauma and neurodegenerative diseases.** Society for Biomaterials Conference, Atlanta, GA, 2018.

Van Drunen, R., Tsai, E., Jimenez-Vergara, A., and Munoz-Pinto, D.: **Fabrication and characterization of multi interpenetrating networks to create a 3D model to study neural trauma and neurodegenerative diseases.** Trinity University Summer Undergraduate Program, San Antonio, TX, 2017.

Van Drunen, R., Lewis, and M., Hildebrandt, M.: **Effects of Doxorubicin on cardiomyocytes derived from IPSCs**. MD Anderson Summer Undergraduate Research Program, Houston, TX, 2016.

Sriraman, H., Broda, C., Chau, E., Van Drunen, R., Sampaio, L., Gobin, A., and Taylor, D**. dECM Endothelization to Create a Non-Thrombogenic Interface of an Engineered Vascular Structure.** Biomedical Engineering Conference, Minneapolis, MN, 2016.